AI-Powered Visual Defect Detection System

Introduction

Quality inspection is an essential process in manufacturing industries to ensure that products meet required standards. Traditional manual inspection methods are time-consuming and prone to human error. With advancements in Artificial Intelligence and Computer Vision, automated defect detection systems can identify surface defects accurately and efficiently. This project focuses on developing an AI-based visual defect detection system capable of classifying different types of steel surface defects using deep learning techniques.

Abstract

The objective of this project is to develop an automated visual inspection system for detecting and classifying steel surface defects. The NEU Surface Defect Database was used for training and evaluation. Initially, a custom Convolutional Neural Network (CNN) was developed to establish baseline performance. To improve accuracy, Transfer Learning was implemented using a pretrained ResNet18 model. The final model achieved a validation accuracy of **99.17%** and was deployed using Streamlit, allowing users to upload images and obtain real-time defect predictions along with confidence scores.

Tools Used

- Python
- PyTorch
- Torchvision
- NumPy
- Pandas
- Matplotlib
- Scikit-learn
- Streamlit
- Jupyter Notebook
- Visual Studio Code

Steps Involved in Building the Project

1. Dataset Collection and Exploration

The NEU Surface Defect Database was used for this project. The dataset contains six defect categories: Crazing, Inclusion, Patches, Pitted Surface, Rolled-in Scale, and Scratches. Dataset exploration was performed to understand class distribution, image properties, and sample defect patterns.

2. Data Preprocessing

Images were resized and converted into tensor format suitable for deep learning models. Training and validation datasets were prepared using PyTorch Data Loaders.

3. Baseline CNN Model

A custom CNN architecture consisting of convolutional, pooling, and fully connected layers was developed. The model achieved a validation accuracy of **74.44%,** indicating limitations in generalization.

4. Transfer Learning with ResNet18

A pre-trained ResNet18 model was fine-tuned for six-class defect classification. Transfer learning significantly improved performance by utilizing features learned from large-scale image datasets. The final model achieved a validation accuracy of **99.17%.**

5. Model Evaluation

The model was evaluated using Accuracy, Precision, Recall, F1-Score, Classification Report, and Confusion Matrix. Results showed high performance across all defect categories with minimal misclassification.

6. Deployment

The trained ResNet18 model was deployed using Streamlit. The web application allows users to upload a steel surface image and receive the predicted defect category along with a confidence score.

Conclusion

This project successfully demonstrates the application of deep learning for automated visual defect detection. A comparison between a custom CNN and a Transfer Learning-based ResNet18 model showed that transfer learning provides significantly better classification performance. The final model achieved a validation accuracy of **99.17%** and was successfully deployed through a Streamlit application. The developed system can serve as a foundation for AI-driven industrial quality inspection solutions.